What happened in our v4 real-robot test, and why

Machine Minds · FANUC ER-4iA cell · 27 October 2026, 16:48–17:18 · written from the robot's own logs and camera images

> **The short version:** the robot did **not** follow positions decided in advance. It only works from what the camera sees at that moment. It failed for two other reasons:
>
> - **The real arm moves about 5× slower than we planned for.** It kept stopping short, and the gripper closed before it reached the object.
> - **Our model missed most of the real objects.** It was trained only on computer-rendered images and had never seen a real photo.

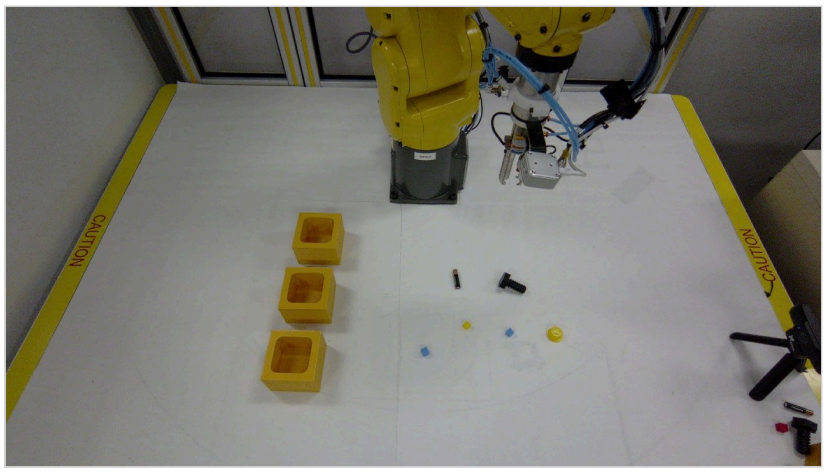
1. "It knew the positions before we even placed the objects." Why it looked like that

The positions on the screen before anything was placed were **our own placement file** (`placement.yaml`). We sent it with the submission so the operator knows where to put each object in each rollout. The organizers' screen shows it as the setup plan. It is not something our model predicted.

Our robot program never reads that file. While running, it only gets three things:

- the live picture from the table camera;
- the arm's joint angles and whether the gripper is open or closed;
- where the arm's fingertip currently is.

The logs back this up: every time the arm went for an object, that object had just been found in a fresh camera picture.



What the robot saw at the start of trial t01: three yellow bins on the left, the objects in the middle, and the arm parked at the top right. The robot found all 3 bins and decided which bin gets which object type. It found the cap and one AAA cell, saw one cube only faintly, and missed the rest. Here the "screw" was a large hex bolt.

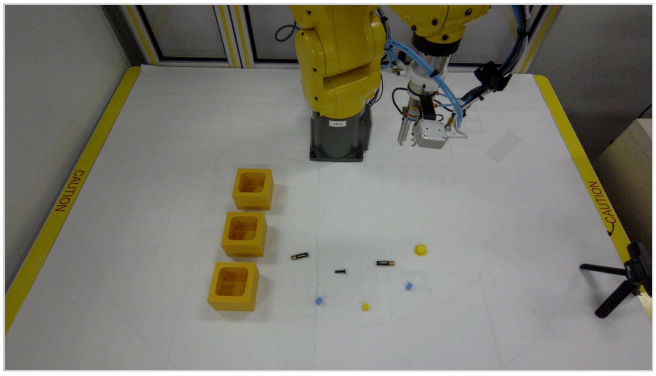
2. Why the arm kept going to the same place

At the start of every trial, our program first moves the arm out of the way (we call it **"park"**) so the arm doesn't block the camera. Only then does it look at the table and pick.

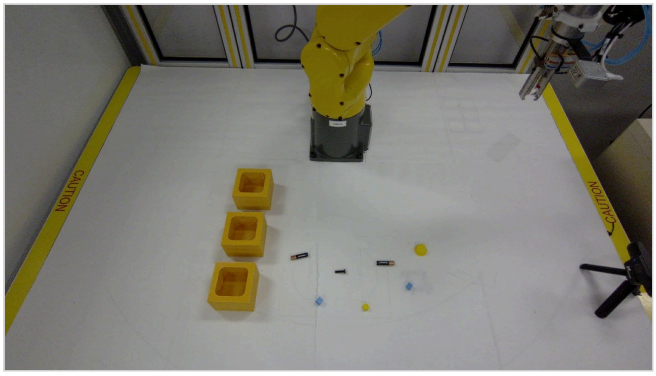
The problem was speed:

- We planned the moves for about **15 cm per second**, but the real arm only managed about **2.5–3 cm per second**, roughly 5× slower.
- The arm stops when its list of moves runs out, even if it hasn't arrived. So each "park" only got it 6–8 cm of the way.
- The program then saw the arm still wasn't parked and sent "park" again, about six times in a row in every trial.

From the outside, the arm moving to the same corner over and over looks like it's visiting fixed positions. It was really the same park move being repeated.



Trial t03, start: the arm is in the middle, above the table.

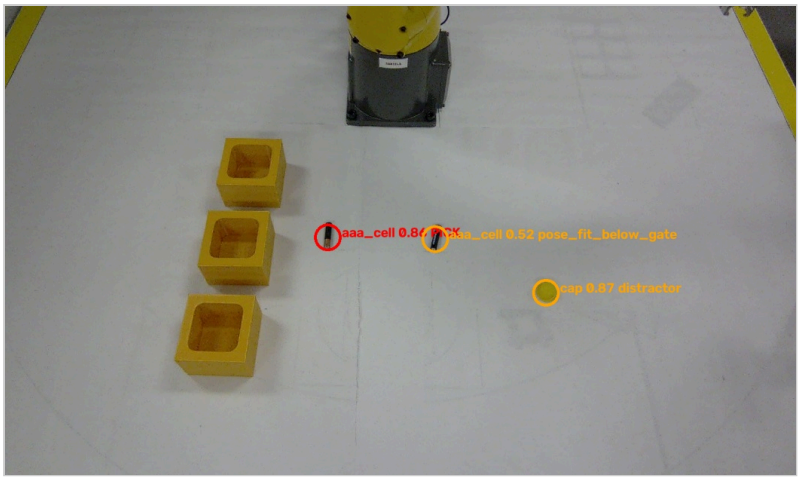


Five "park" moves later: the arm has only now reached the park spot at the top right.

3. Why it missed the objects it did try to pick

A pick is one long list of moves: go above the object, go down, close the gripper, lift, carry to the bin, open, come back. The "close the gripper" step happens at a **set time** in that list, not when the arm actually arrives.

Because the arm was about 5× slower than planned, it was still on its way when "close" came. So **the gripper closed in the air, before reaching the object**, and the arm carried nothing to the bin. This happened to the AAA cell in trial t03. The next picture showed the cell still there, so the robot correctly tried again, and missed again for the same reason.



Trial t03, act 15. The red circle is the AAA cell the robot went for, twice; both times the gripper closed too early. The orange circles are things it saw but chose not to pick: a second AAA cell it wasn't sure about, and the cap, which it should leave alone.

> **One thing that did check out:** the arm's position as reported by the real controller matched our digital twin almost exactly, to about a hundredth of a millimetre. So our coordinates are right; the problem is the timing.

4. Why it didn't see most of the objects

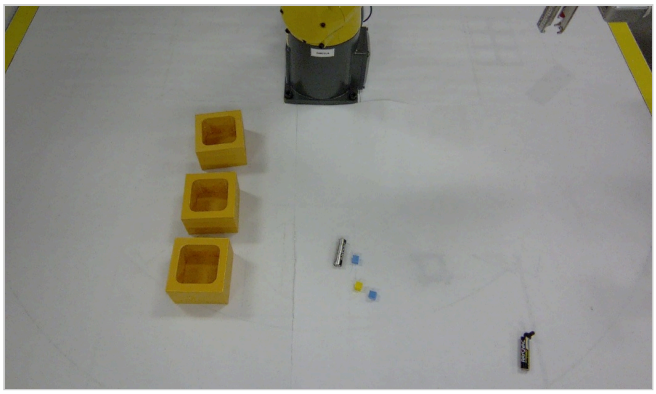
Our object-detection model (YOLO) was trained only on **computer-rendered images** of the digital twin. This was the first time it saw real camera photos, and it struggled:

- **Bins:** found almost every time (in 175 of 182 pictures). These worked well.
- **Cap:** found reliably and correctly left alone.
- **AAA cells:** mostly not found; only twice was it confident enough to pick one.
- **Cubes:** very small (about 10 pixels wide), usually seen only faintly, so our safety checks refused to pick them.
- **Screw:** sometimes seen but always refused as uncertain. In the first trial it was a big hex bolt, which the model was never trained on.
- **Glass tube:** never found.

Our safety checks did their job: the robot didn't grab things it wasn't sure about. But since almost nothing passed them, the robot spent most of each trial holding still, waiting to see something it was sure of.



Trial t05: two AAA cells, two cubes and the glass tube are on the table, but the robot saw only one cube (orange circle) and wasn't sure enough to pick it.



Trial t06: AAA cells and cubes are clearly on the table, and the robot detected nothing at all.

5. The trials, in short

- **t00:** ended half a second after it started, before our program did anything.
- **t01, first try:** parked, then got stuck waiting for the cell, which then went offline. We restarted the client.
- **t01, re-run:** parked, found the 3 bins, picked their roles, and started its first real pick (an AAA cell). The trial ended 17 seconds in, during that pick.
- **t02:** six parks, then held still; nothing it was sure about.
- **t03:** six parks, then tried the AAA cell twice. The gripper closed too early both times.
- **t04, t05 and t06:** six parks each, then held still; it saw almost nothing it was sure about.

6. What we'll fix

- **Wait for the arm.** Slow our moves down to the arm's real speed, and close or open the gripper only once the arm has actually arrived (the controller tells us where it is).
- **Teach the model real photos.** We saved 182 real pictures from this test. Labelling them and retraining will close most of the "can't see it" gap.
- **Use the right objects.** Use the countersunk screws the model was trained on (not the hex bolt), and photograph the real AAA cells, cubes and tube so the twin looks more like them.
- **Park once.** Wait until the arm has reached the park spot, so each trial doesn't start with six park moves.

Detail for the team: the full logs and all 182 camera pictures are in `v4\runs\real\20261027_165715\`.